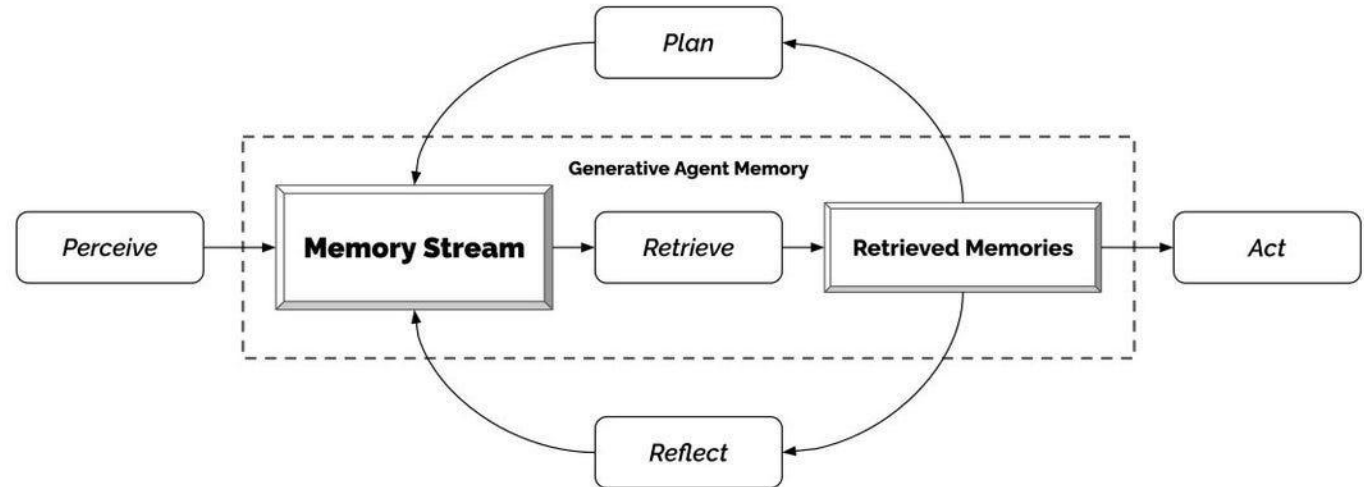
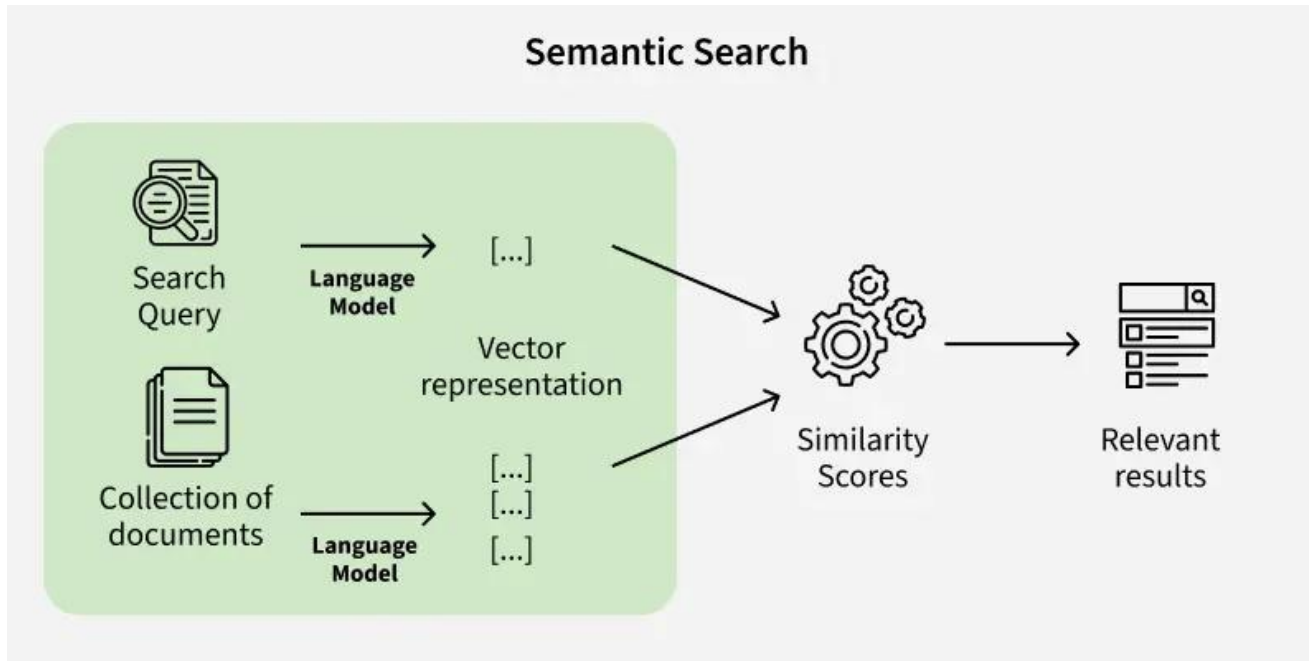


Vector Databases: Memory for Your AI

- Unlike traditional SQL databases that store text or numbers in rows, **Vector Databases** store data as high-dimensional embeddings.
- Tools like **ChromaDB** (local/open-source) and **Pinecone** (cloud-scale) allow for lightning-fast retrieval of related data.
- This is the "hard drive" where you store your local PDFs, spreadsheets, and private documents.

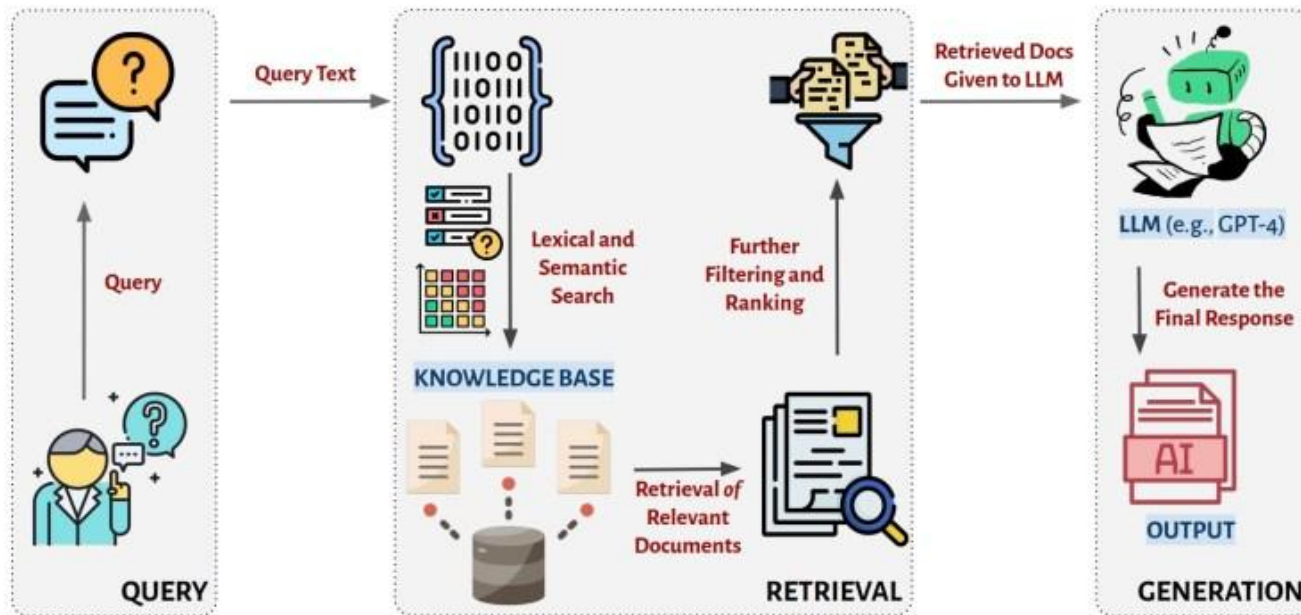


Semantic Search: Finding Meaning, Not Just Words



- Traditional search looks for exact keywords. **Semantic Search** uses embeddings to find the "relevant" page based on the *meaning* of the user's query.
- By converting the user's question into a vector, the system can find the exact paragraph in a 500-page PDF that contains the answer.
- This step acts as a "filter" to ensure the LLM only reads the necessary information.

RAG Pipelines: Eliminating Hallucinations



- A **RAG Pipeline** follows a three-step process: **Retrieve** (find context), **Augment** (add context to the prompt), and **Generate** (ask the LLM).
- This "Chat with your Data" approach ensures the LLM bases its answers on factual local data rather than its own internal training memory.
- It is the industry standard for creating reliable, private, and up-to-date AI assistants.